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Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Collection Classes and StringBuffer

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Programme

```
J CourseRegistration.java > CourseRegistration > main(String[])
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 public class CourseRegistration {
    Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<String> courses = new ArrayList<>();
8         int choice;
9
10        do {
11            System.out.println("\n----- COURSE REGISTRATION -----");
12            System.out.println("1. Register Course");
13            System.out.println("2. Remove Course");
14            System.out.println("3. View Registered Courses");
15            System.out.println("4. Exit");
16            System.out.print("Enter your choice: ");
17            choice = sc.nextInt();
18            sc.nextLine();
19
20            switch (choice) {
21                case 1:
22                    System.out.print("Enter Course Name: ");
23                    String course = sc.nextLine();
24                    courses.add(course);
25                    System.out.println("Course Registered.");
26                    break;
27
28                case 2:
29                    if (courses.isEmpty()) {
30                        System.out.println("No Registered Courses.");
31                    } else {
32                        System.out.print("Enter Course Number to Remove: ");
33                        int index = sc.nextInt();
34                        if (index >= 1 && index <= courses.size()) {
35                            courses.remove(index - 1);
36                            System.out.println("Course Removed.");
37                        } else {
38                            System.out.println("Invalid Course Number.");
39                        }
40                    }
41                    break;

```

```

43
44
45     case 3:
46         StringBuffer sb = new StringBuffer();
47         sb.append(str: "\nRegistered Courses:\n");
48         if (courses.isEmpty()) {
49             sb.append(str: "No Courses Registered.");
50         } else {
51             for (int i = 0; i < courses.size(); i++) {
52                 sb.append((i + 1) + ". " + courses.get(i) + "\n");
53             }
54         }
55         System.out.println(sb);
56         break;
57
58     case 4:
59         System.out.println(x: "Exiting...");
60         break;
61
62     default:
63         System.out.println(x: "Invalid Choice.");
64
65 } while (choice != 4);
66
67 sc.close();
68 }
69 }

```

J TodoList.java >  TodoList

```

1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class TodoList {
5      Run | Debug
6      public static void main(String[] args) {
7          Scanner sc = new Scanner(System.in);
8          ArrayList<String> tasks = new ArrayList<>();
9          int choice;
10
11      do {
12          System.out.println(x: "\n----- TO-DO LIST -----");
13          System.out.println(x: "1. Add Task");
14          System.out.println(x: "2. Remove Task");
15          System.out.println(x: "3. View Tasks");
16          System.out.println(x: "4. Exit");
17          System.out.print(s: "Enter your choice: ");
18          choice = sc.nextInt();
19          sc.nextLine();
20
21          switch (choice) {
22              case 1:
23                  System.out.print(s: "Enter Task: ");
24                  String task = sc.nextLine();
25                  tasks.add(task);
26                  System.out.println(x: "Task Added.");
27                  break;
28
29              case 2:
30                  if (tasks.isEmpty()) {
31                      System.out.println(x: "No Tasks Available.");
32                  } else {
33                      System.out.print(s: "Enter Task Number to Remove: ");
34                      int index = sc.nextInt();
35                      if (index >= 1 && index <= tasks.size()) {
36                          tasks.remove(index - 1);
37                          System.out.println(x: "Task Removed.");
38                      } else {
39                          System.out.println(x: "Invalid Task Number.");
40                      }
41                  }
42                  break;

```

```

43         case 3:
44             StringBuffer sb = new StringBuffer();
45             sb.append(str: "\nYour Tasks:\n");
46             if (tasks.isEmpty()) {
47                 sb.append(str: "No Tasks Available.");
48             } else {
49                 for (int i = 0; i < tasks.size(); i++) {
50                     sb.append((i + 1) + ". " + tasks.get(i) + "\n");
51                 }
52             }
53             System.out.println(sb);
54             break;
55
56         case 4:
57             System.out.println(x: "Exiting...");
58             break;
59
60         default:
61             System.out.println(x: "Invalid Choice.");
62     }
63
64     } while (choice != 4);
65
66     sc.close();
67 }
68 }

```

Output

```

----- COURSE REGISTRATION -----
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 1
Enter Course Name: Btech CSE
Course Registered.

```

```

----- COURSE REGISTRATION -----
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 4
Exiting...

```

```

----- TO-DO LIST -----
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 1
Enter Task: Drink Water
Task Added.

```

```

----- TO-DO LIST -----
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 4
Exiting...

```

Conclusion

This assignment successfully demonstrated the use of ArrayList, Vector and StringBuffer in java for dynamic data storage and efficient string manipulation. The developed applications provided functionalities to add, remove and display course registrations and to-do list tasks.